



Introduction to Data Science

Final Project Phase 1

Instructors: **Dr. Yaghoobzadeh**
Dr. Razavi

TAs: **Mina Shirazi, Mostafa Kermani Nia,**
Mehrad Livian, Mohammad Amanlou,
Mohammad Reza Mohammad Hashemi,
Yasaman Amou Jafary

In this project, you will complete a full data science project and apply the concepts and principles you have learned theoretically in practice. The project will be carried out step by step.

In this phase, which accounts for **10% of your final project grade**, each team must choose the dataset they intend to analyze and investigate. You are required to submit the specifications of your selected dataset by the announced deadline and wait for approval from the teaching assistants. Each team must send an email to [this](#) address.

Your email must include the following information:

1. **Dataset name:** Please mention whether the dataset has been collected/crawled by your team.
2. **Source URL:** Provide the URL of the website or platform from which the data has been collected.
3. **Sample data link:** Include a link to a sample of the extracted data.
4. **Project target:** Clearly explain the goal of your project and what you aim to analyze, predict, classify, or investigate using this dataset.
5. **Suggested mentors:** Please select three preferred mentors from the list below and indicate your priorities from 1 to 3, where Priority 1 is your most preferred mentor.

Available mentors:

- Mina Shirazi
- Mostafa Kermani Nia
- Mehrad Livian
- Mohammad Amanlou
- Mohammad Reza Mohammad Hashemi
- Yasaman Amou Jafary

After the teaching assistants review and approve your selected dataset, a mentor will be assigned to your team as the main project guide. Your mentor will support your team throughout the different stages of the project and provide guidance to help you execute the project correctly. You are encouraged to ask your mentor or teaching assistant any questions you may have during the project.

After your dataset has been reviewed and approved by the teaching assistants, you should proceed to fully collect and prepare your final dataset for the next phase of the project. Make sure the data is complete, clean, well-structured, and ready for analysis so that your team can effectively continue with the upcoming stages of the project.

If you submit your dataset earlier than other groups, you will receive approval sooner and can proceed with the subsequent stages. Additionally, this will reduce the likelihood of another group selecting a dataset similar to yours. Therefore, please make sure to choose your dataset and email it as soon as possible. The teaching assistants will continuously respond to the datasets emailed.

Important Notes

1. In this project, the data of each group must be unique. Therefore, if two groups submit similar data, only the group that registers the specifications earlier can use the data.
2. Checking and confirming the data by teaching assistants may take several days, so please be patient. If your work isn't checked for more than 3 days, you may notify teaching assistants via email.
3. If your data is not approved by the teaching assistants, you'll have to resubmit the specifications. Therefore, don't delay submitting your dataset specifications until the final days.
4. Teams that collect or crawl their own dataset instead of using an existing ready-made dataset will be eligible for a **bonus of up to 100% of the first phase score. In other words, by doing so, you can receive up to twice the total score allocated to this phase.**
5. To find datasets, you can use websites such as [Kaggle](#) and [Hugging Face](#), or other similar websites. In addition, for datasets related to Iran, you may also explore Iranian or Iran-focused data sources such as Iran Open Data, Data.gov.ir, Statistical Center of Iran, and Iran Data Portal. These platforms can be useful for accessing or collecting data in areas such as economics, population, society, transportation, public services, and other Iran-related topics.

Required Data Specifications

1. The dataset you select should have sufficient complexity to support a meaningful data science project. This complexity can come from working with data types such as images, audio, time series, text, video, signals, or from using methods such as large language model agents. However, complex data is not limited to these formats. You may also choose a structured tabular dataset, as long as it contains rich, informative, and workable features that allow for meaningful analysis, modeling, feature engineering, and insight extraction. The difficulty level of your dataset will be assessed by the team of teaching assistants. Collected datasets are categorized into three levels:
 - a. Rejected: This level indicates that the dataset is too simplistic and not suitable for use in the final project of the course.
 - b. Acceptable: This level signifies that your dataset is deemed acceptable for the course project.
 - c. Advanced: This level indicates that your dataset is considered advanced and will receive a scoring advantage (**a 10% bonus on the overall project grade**).
In general, employing advanced datasets, complex pipelines, sophisticated pre-processing techniques, and higher-level processing methods can earn you up to a 10% bonus score. Moreover, this bonus may be applied again in later phases if your work continues to demonstrate complexity. Please ensure that you take the requirements of future phases into account when selecting your dataset, so that it supports all upcoming tasks. The complexity criteria will be determined by the TAs. If your current dataset does not meet the necessary complexity standards, you will have the opportunity to select a new one.
2. In the final phases of the project, you must implement a data-driven task that leverages the relationships within your dataset. While a common approach might involve predicting one data column using the others—by selecting a target variable and ensuring that a logical and correlated relationship exists—this requirement is intentionally flexible. You are free to choose other tasks, such as developing a recommendation system, generating text, performing image segmentation, or any other application that fits your dataset.
 - Regardless of your chosen approach, your methodology should capitalize on the inherent structure and interconnections within your data. Ensure that your selected task is well-supported by your dataset's characteristics and that your approach is logically justified by the relationships present in the data.

Overview of the Next Phases of the Project

The project consists of three main phases.

Phase 1: Topic Selection and Data Collection

In this phase, teams will focus on defining their project core objectives and gathering their primary datasets.

Phase 2: Data Storage and Processing Pipeline

In the second phase, you are required to store your data in an appropriate environment and establish a proper pipeline, which will be covered in the subsequent lessons. This will enable you to read and process the data safely, allowing you to extract the necessary results. Additionally, during this phase, you should perform some preprocessing tasks and address questions related to the nature of your dataset.

Phase 3: Final Conclusions and Model Development

In the third phase, you will arrive at final conclusions and develop a model to execute your defined task

These phases are designed to guide you systematically through the project, ensuring that you build a solid analytical foundation and derive valuable insights from your data.

Good luck!